

Paper 2 · Answers & Solutions

2026 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	A	D	B	B	C	C	A	B	D	A	C	D	A
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	B	A	D	C	D	B	D	B	A	B	B	D	D	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	A	B	C	C	D	A	D	C	A	A	D	B	C	C

Part 2 · Worked Solutions

Section A

1 C

B-
Indices

1 $3^{-1} + 6^{-1} = \frac{1}{2}$

2 Reciprocal = 2, choose C

2 A

B
Linear Equations

1 $\alpha x = \beta(1 - x) \Rightarrow (\alpha + \beta)x = \beta$

2 $x = \frac{\beta}{\alpha + \beta}$, choose A

3 A

B
Factorization

1 $m^2 - 9n^2 = (m + 3n)(m - 3n);$
 $-5m - 15n = -5(m + 3n)$

2 $= (m + 3n)(m - 3n - 5)$, choose A

4 D

B
Algebraic Fractions

1 Combine: $\frac{2(3x + 4) + 2(3x - 4)}{(3x - 4)(3x + 4)}$

2 $= \frac{12x}{9x^2 - 16}$, choose D

5 B

B
Graphs of Quadratic Functions

1 Opens downward, vertex (4, 25)

2 Vertex above x -axis, opens down $\Rightarrow$ cuts x -axis, choose B

6 B

A-
Graphs of Straight Lines

1 From slopes and intercepts of L_1, L_2

2 I and II hold, III not necessarily, choose B

7 C

B
Functions

1 $f(2m + 1) = 2(2m + 1)^2 + (2m + 1) - 3$

2 $= 8m^2 + 10m$, choose C

8 C

B+
Remainder Theorem

1 Divisible by

$$x - 1 \Rightarrow g(1) = 1 + a + b = 0$$

2 $g(-1) = 1 + a + b = 0$; remainder = 0, choose C

9 A

B+
Compound Interest

1 $A = 80000(1 + \frac{0.03}{12})^{24}$

2 Interest ≈ 4941 , choose A

10 B

A-
Ratios

1 $b = \frac{3a}{5}, c = \frac{7a}{2}$

2 $= \frac{26}{115}$, choose B

11 D

A-
Variation

1 $w = \frac{k\sqrt{u}}{v^2} \Rightarrow k = \frac{wv^2}{\sqrt{u}}$

2 Square: $\frac{w^2v^4}{u}$ constant, choose D

12 A

B+
Sequences

1 $a_5 = a_3 + a_4, a_6 = a_4 + a_5$

2 $a_4 = 36, a_5 = 59$, choose A

13 C

B+
Inequalities

1 $\frac{2x+1}{3} \geq x - 3 \Rightarrow x \leq 10$

2 $4x + 9 < 1 \Rightarrow x < -2$; union $x \leq 10$, choose C

14 D

A-
Accuracy of Measurement

1 Each length correct to nearest cm, max error 0.5

2 Area range $21.25 \leq x < 42.25$, choose D

15

A

B+
Volume of Solids

- 1 Triangular prism: (triangle area) $\times$ length
- 2 Volume = **360** cm³, choose A

16

B

A
Ratios of Areas

- 1 $BE : EC = 5 : 3$, $[ABF] = 120$; use affine area ratios
- 2 $[CDFE] = 237$ cm², choose B

17

B

A
Sectors

- 1 $OE = 18$, $AD = 48$; radius OA
- 2 Sector $OBC = 75\pi$, choose B

18

A

A
Rhombuses

- 1 $AE = AF$,
 $\angle BAD = 110^\circ \Rightarrow \angle AEF = 35^\circ$
- 2 With $\angle ECF = 42^\circ$, $\angle BEC = 76^\circ$, choose A

19

D

A
Regular Polygons

- 1 Regular pentagon, interior **108°**
- 2 I, II and III all true, choose D

20

C

A
Squares

- 1 Square, $BE = 8$, $EF = 10 \Rightarrow BF = 6$
- 2 By similar triangles $DF = 30$, choose C

21

D

A
Trapeziums and Trigonometry

- 1 E, F trisect AB ; $\angle ABC = \angle BAD = 90^\circ$
- 2 II and III must hold, choose D

22

B

A
Properties of Circles

- 1 $BD = DE \Rightarrow \angle DBE = \angle E$
- 2 $\angle CED = 28^\circ$, choose B

23

D

B
Symmetry

- 1 Eight identical squares in a symmetric figure
- 2 Rotational symmetry order = **4**, choose D

24

B

A-
Polar Coordinates

- 1 Adjacent polar angles differ by **90°**
- 2 Perimeter = **70**, choose B

25

A

A-
Locus

- 1 L_2 is parallel to L_1
- 2 Locus is the mid-line: $24x - 8y + 17 = 0$, choose A

26

B

A
Straight Lines and Area

- 1 L_1 meets y -axis at $(0, 12)$; $L_2 \perp L_1$ there
- 2 Triangle area = **150**, choose B

27

B

A-
Equations of Circles

- 1 Centre $(3, -1)$, $r^2 = \frac{44}{5}$
- 2 Circumference $\approx 18.7 < 20$, choose B

28

D

A-
Probability

- 1 $C_2^7 = 21$ ways
- 2 5 favourable, $P = \frac{5}{21}$, choose D

29

D

B+
Mean

- 1 Total = **1320**; six = **648**
- 2 Remaining mean = $\frac{672}{4} = 168$, choose D

30

C

A
Stem-and-Leaf Diagrams

- 1 $IQR \leq 25$ constrains a, b
- 2 I and II must hold, choose C

31

A

A
Transformations of Graphs

- 1 $y = f(-x + 11)$ is a horizontal reflection/translation
- 2 Consistent with the figure, choose A

32

A

A
Graphs of Logarithmic Functions

- 1 From positions of $y = \log_a x, \log_b x$
- 2 II and III hold, choose A

33

B

A
Logarithms (Linearisation)

- 1 Slope $= \frac{10 - 4}{5 - 1} = \frac{3}{2} = k$
- 2 $k = \frac{3}{2}$, choose B

34

C

A
Linear Programming

- 1 Evaluate $5x + 6y + 206$ at vertices
- 2 Least value $= 150$, choose C

35

C

A
Sequences

- 1 $a_n = 12n - 7$ ($n \geq 2$), $a_1 = 5$
- 2 Only II holds, choose C

36

D

A
Quadratic Equations

- 1 m, n roots of $2t^2 + 7t - 15 = 0$: sum $-\frac{7}{2}$, product $-\frac{15}{2}$
- 2 $= -9$, choose D

37

A

A-
Complex Numbers

- 1 Multiply by $\frac{1 + 2i}{1 + 2i}$
- 2 Real part $= \frac{-15}{5} = -3$, choose A

38

D

A
Trigonometric Equations

- 1 $6 \cos^2 x - \cos x - 5 = 0 \Rightarrow \cos x = 1$ or $-\frac{5}{6}$
- 2 $\cos x = -\frac{5}{6}$ gives 2 roots, choose D

39

C

A
Circles and Tangents

- 1 Tangent-chord angles with $AB = CD$
- 2 $\angle ABC = 78^\circ$, choose C

40

A

A+
Centres of Triangle

- 1 $P = (\frac{a}{2}, 0), Q = (0, \frac{a}{5}), R$ on y -axis
- 2 Orthocentre $x = 10 \Rightarrow R_y = 4$, choose A

41

A

A+
Angles in 3-D

- 1 $DX = 9, EX = 4; BX$ to plane $ABGF$
- 2 $\cos \theta = \frac{12}{13}$, choose A

42

D

A
Combinations

- 1 Same gender: $C_3^{14} + C_3^{15}$
- 2 $= 364 + 455 = 819$, choose D

43

B

A
Probability

- 1 John throws on odd turns; $P('5') = \frac{1}{6}$, $P(\text{stop}) = \frac{1}{3}$
- 2 Sum of geometric series $= \frac{3}{10}$, choose B

44

C

A
Standard Scores

- 1 $-3 = \frac{50 - 74}{sd} \Rightarrow sd = 8$
- 2 Susan: $\frac{90 - 74}{8} = 2$, choose C

45

C

A-
Variance

- 1 Variance depends on spacing, not position; same common difference
- 2 Variance of last 6 terms $= 16$, choose C

